INCH-POUND

MIL-M-28787/218A(SH)
AMENDMENT 1
19 November 1990

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

FILTER, LOW PASS, ACTIVE, ANALOG, KEY CODE EEE

This amendment forms a part of Military Specification MIL-M-28787/218A(EC), dated 15 June 1982, and is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 5

TABLE I: Delete and substitute new table I as printed on page 3 of this amendment

PAGE 6

TABLE 1: Test condition G10: Delete "See note 6, figure 9" and substitute "See note 6, figure 10".

PAGE 7

TABLE I: Delete and substitute new table I as printed on page 4 of this amendment

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4.5.1, fifth line: Delete "C through L" and substitute "C, E, G1-G10, H1, H2, H4, and J through L".

4.5.3, third and sixth line: Delete "C through L" and substitute "C, E, G1-G10, H1, H2, H4, and J through L".

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TABLE VI: Test condition G10: Delete "See note 6, figure 9" and substitute "See note 6, figure 10".

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TABLE VI: Test condition G20: Delete "See note 6, figure 9" and substitute "See note 6, figure 10".

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TABLE VII: Delete and substitute new table VII as printed on page 5 of this amendment.

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TABLE VIII: Delete and substitute new table VIII as printed on page 6 of this amendment.

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FIGURE 12. Delete and substitute new figure 12 as printed on page 7 of this amendment.

Preparing activity:
Navy - SH

(Project 5963-N780)

Agent: NW

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TABLE I. Module electrical requirements.

Test cond.	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25 ± 5°C ambient temperature				Operating temperature 0 to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1	Isolation, used to unused	II	X	10	∞	1	∞	10	∞	1	∞	MΩ
A2	Isolation, frame	II	S	10	∞	1	∞	10	∞	1	∞	MΩ
B1, B2	Output short circuit duration	III	Q	10	---	10	---	10	---	10	---	s
C1	DC offset voltage	III	X	-20	20	-20	20	-20	20	-20	20	mV
C2	DC offset voltage	III	X	-15	15	-15	15	-17	17	-17	17	mV
D1-D2	DC gain	III	Q	0.95	1.05	0.95	1.05	0.95	1.05	0.95	1.05	V
E1-E2	Output voltage swing	IV	S	7.0	---	7.0	---	7.0	---	7.0	---	V
F1, F2	Power supply current	V	Q	1.0	8.5	1.0	8.5	1.0	9.0	1.0	9.0	mA
G1	Gain versus frequency at 9600 Hz	VI	X	-0.5	1.5	-0.5	1.5	-0.5	1.5	-0.5	1.5	dB
G2	Gain versus frequency at 20 to 7000 Hz	VI	X	-2.0	0.5	-2.0	0.5	-2.0	0.5	-2.0	0.5	dB ref G1
G3	Gain versus frequency at 7600 Hz	VI	X	-0.3	0.3	-0.3	0.3	-0.3	0.3	-0.3	0.3	dB ref G1
G4	Gain versus frequency at 9540 Hz	VI	X	-0.25	0.25	-0.25	0.25	-0.25	0.25	-0.25	0.25	dB ref G1
G5	Gain versus frequency check at 9660 Hz	VI	X	-0.25	0.25	-0.25	0.25	-0.25	0.25	-0.25	0.25	db ref G1
G6	Gain versus frequency check at 11600 Hz	VI	X	-0.3	0.3	-0.3	0.3	-0.3	0.3	-0.3	0.3	dB ref G1
G7	Gain versus frequency check at 15 kHz	VI	X	-0.5	0.5	-0.5	0.5	-0.5	0.5	-0.5	0.5	dB ref G1

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25 ± 5°C ambient temperature				Operating temperature 0 to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
G20	Gain versus frequency check 16 to 100 kHz	VI	Q	◄----------See note 6, figure 10----------►								
H1	Phase shift (cw)	VII	X	100	120	100	120	100	120	100	120	Degree
H2	Phase shift (ccw)	VII	X	240	260	240	260	240	260	240	260	Degree
H3	Phase shift	VII	Q	260	275	260	275	260	275	260	275	Degree
H4	Phase shift settability	VII	S	-0.5	0.5	-0.5	0.5	-0.5	0.5	-0.5	0.5	Degree
H5	Phase shift setting under pressure	VII	Q	-0.5	0.5	-0.5	0.5	-0.5	0.5	-0.5	0.5	Degree
H6	Phase shift average temperature coefficient check	VII	Q	--	--	--	--	0.0	0.05	0.0	0.05	Degree/C
J1, J2	Distortion check	VIII	S	0	2	0	2	0	2	0	2	%
J3, J4	2nd harmonic	VIII	X	-∞	-48	-∞	-48	-∞	-48	-∞	-48	dB
K1, K2	Power supply transient rise time	IX	S	27.5	77.5	27.5	77.5	27.5	77.5	27.5	77.5	µS
L	Continuity	X	S	0	1	0	1	0	1	0	1	Ω

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (one percent or two modules from each inspection lot, whichever is greater) at 25°C

3/ "Q" parameters shall be tested during qualification testing only.

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TABLE VII. Phase shift test.

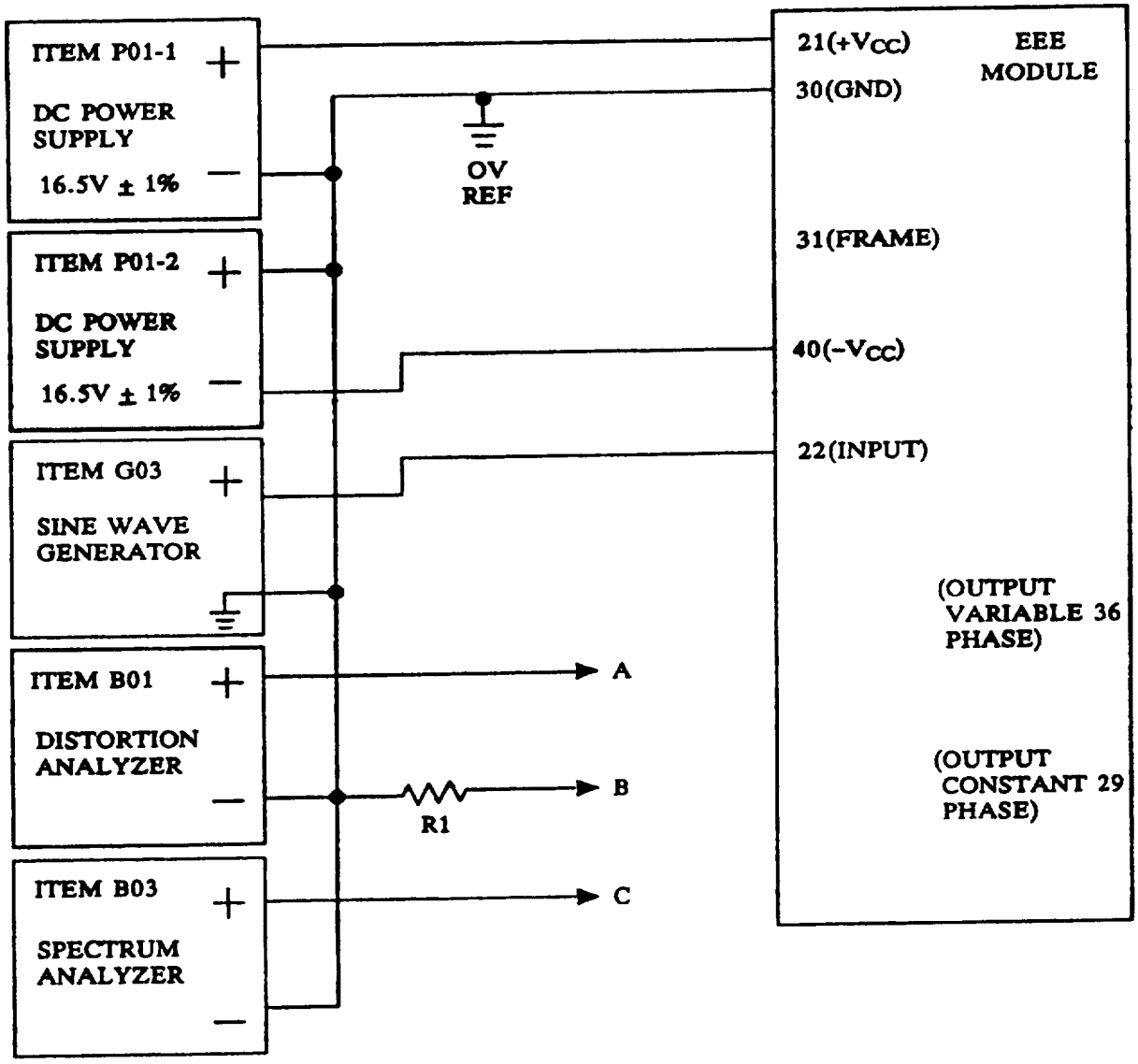
Test condition	Parameter	Connect	Initial limits 25 ± 5°C, ambient		Unit	Test figure
		A to pin	Min	Max		
H1	Phase shift (potentiometer fully clockwise)	36	100	120	Degree	11
H2	Phase shift (potentiometer fully counterclockwise)	36	240	260	Degree	11
H3	Phase shift	29	260	275	Degree	11
H4	Phase shift (settability)	36	-0.5	0.5	Degree	11
H5	Phase shift (setting under pressure)	—	-0.5	0.5	Degree	11
H6	Phase shift (average temperature coefficient) check	—	—	—	Degree	11

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TABLE VIII. Distortion and 2nd harmonic test.

Test condition	Parameter	Connect		Initial limits 25 ± 5°C, ambient		Unit	Test figure
		B to pin	C to pin	Min	Max		
J1	Distortion check	36	Open	0	2	%	12
J2	Distortion check	29	Open	0	2	%	12
J3	2nd harmonic	Open	36	-∞	-48	dB	12
J4	2nd harmonic	Open	29	-∞	-48	dB	12

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NOTES:
1. R1 = 2kΩ ± 1%, 1/8W.
2. Set the input from the item G03 to 4.0/Vrms ± 1% at 9600 Hz ± 10%.
3. Test condition J1. Connect point A to pin 22, and measure input distortion, D_i. Then connect point A to point B and measure output distortion, D_0. Then calculate distortion, D_i with the formula:

$$D = D_0 - D_i$$

4. Repeat the above procedure for test condition J2.
5. For test conditions J3 and J4, measure the magnitude of the 2nd harmonic present at 19,200 Hz ± 20%.

FIGURE 12 Distortion and 2nd harmonic test circuit for table VIII.

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